Name : Thube Shivam

PRN : 202501050036

2.1.1. List operations

17:40

Write a Python program that implements a menu-driven interface for managing a list of integers. The program should have the following menu options:

1. Add
2. Remove
3. Display
4. Quit

The program should repeatedly prompt the user to enter a choice from the menu. Depending on the choice selected, the program should perform the following actions: Add: Prompts the user to enter an integer and add it to the integer list. If the input is not a valid integer, display "Invalid input".

Remove: Prompts the user to enter an integer to remove from the list. If the integer is found in the list, remove it; otherwise, display "Element not found". If the list is empty, display "List is empty".

Display: Displays the current list of integers. If the list is empty, display "List is empty". Quit: Exits the program.

The program should handle invalid menu choices by displaying "Invalid choice".Ensure that the program continues to prompt the user until they choose to quit (option 4).

Test case 1

```
1.· Add ⏎
2.·Remove⏎
3.·Display⏎
4.·Quit⏎
Enter·choice:·1
Integer:·11
Lis t·a ft er· add in g:·⏎[1 1]
```

```
1.·Add↵
2.·Remove↵
3.·Display↵
4.·Quit↵
Enter·choice:·1
Integer:·25
List·after·adding:·[11,·25]↵
1.·Add↵
2.·Remove↵
3.·Display↵
4.·Quit↵
Enter·choice:·2
Integer:·26
Element·not·found↵
1.·Add↵
2.·Remove↵
3.·Display↵
4.·Quit↵
Enter·choice:·2
Integer:·11
List·after·removing:·[25]↵
1.·Add↵
2.·Remove↵
3.·Display↵
4.·Quit↵
```

```
Enter choice: 2
Integer: 25
List after removing : []
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 3
List is empty
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 2
List is empty
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 8
Invalid choice
1. Add
2. Remove
3. Display
4. Quit
```

```
Enter·choice:·4
```

Test case 2

```
1.· Add ⏎
2.·Remove⏎
3.·Display⏎
4.·Quit⏎
Enter·choice:·1
Integer:·1
List·after·adding:·[1]⏎
1.·Add⏎
2.·Remove⏎
3.·Display⏎
4.·Quit⏎
Enter·choice:·1
Integer:·2
List·after·adding:·[1,·2]⏎
1.·Add⏎
2.·Remove⏎
3.·Display⏎
4.·Quit⏎
Enter·choice:·1
Integer:·3
Lis t·a ft er· add in g:· [1⏎·2 ,· 3]
1.·Add⏎
2.·Remove⏎
```

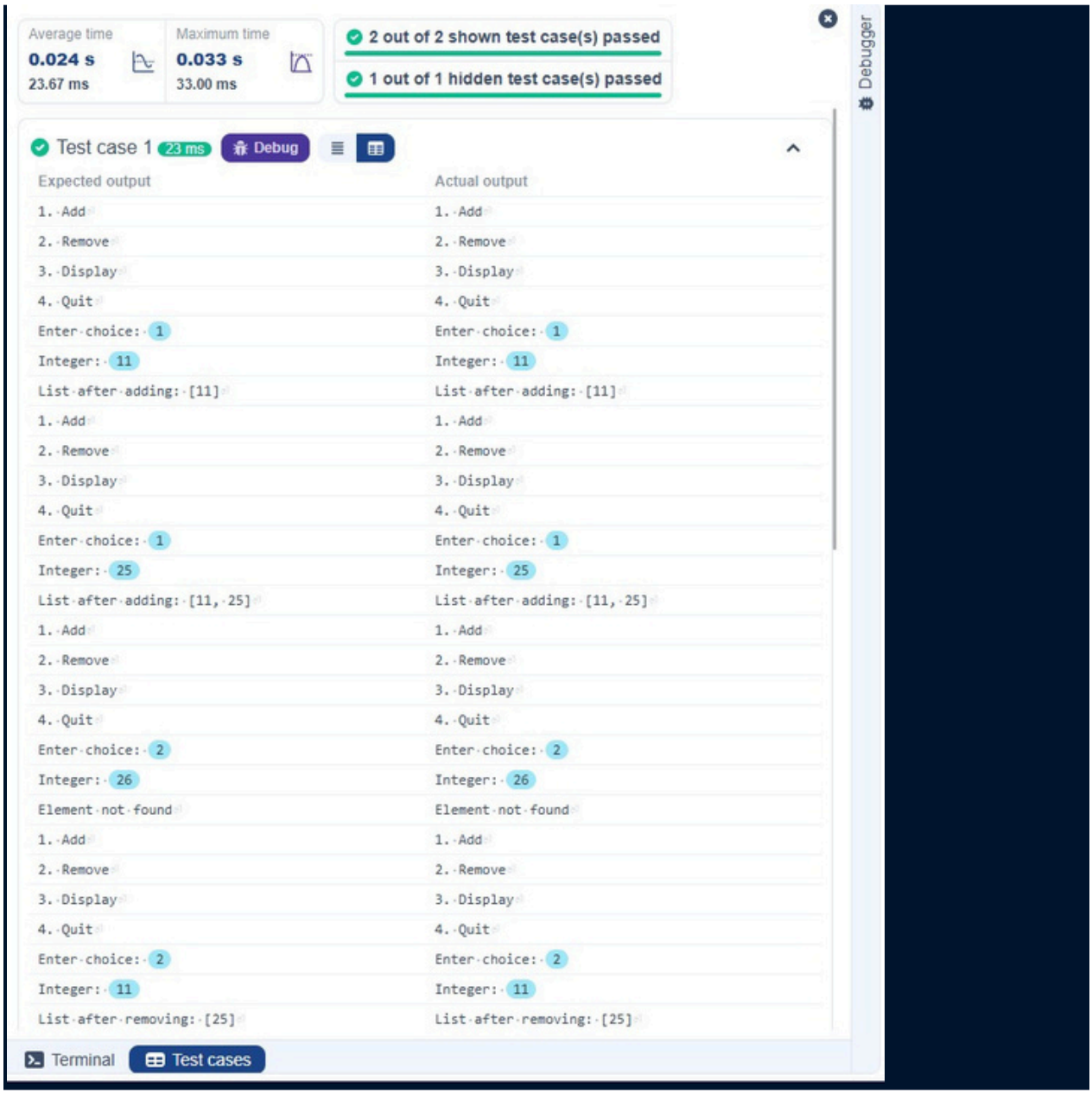
```
3.·Display↵
4.·Quit↵
Enter·choice:·1
Integer:·4
List·after·adding:·[1,·2,·3,·4]↵
1.·Add↵
2.·Remove↵
3.·Display↵
4.·Quit↵
Enter·choice:·2
Integer:·2
List·after·removing:·[1,·3,·4]↵
1.·Add↵
2.·Remove↵
3.·Display↵
4.·Quit↵
Enter·choice:·2
Integer:·6
Element·not·found↵
1.·Add↵
2.·Remove↵
3.·Display↵
4.·Quit↵
Enter·choice:·3
[1,·3,·4]↵
```

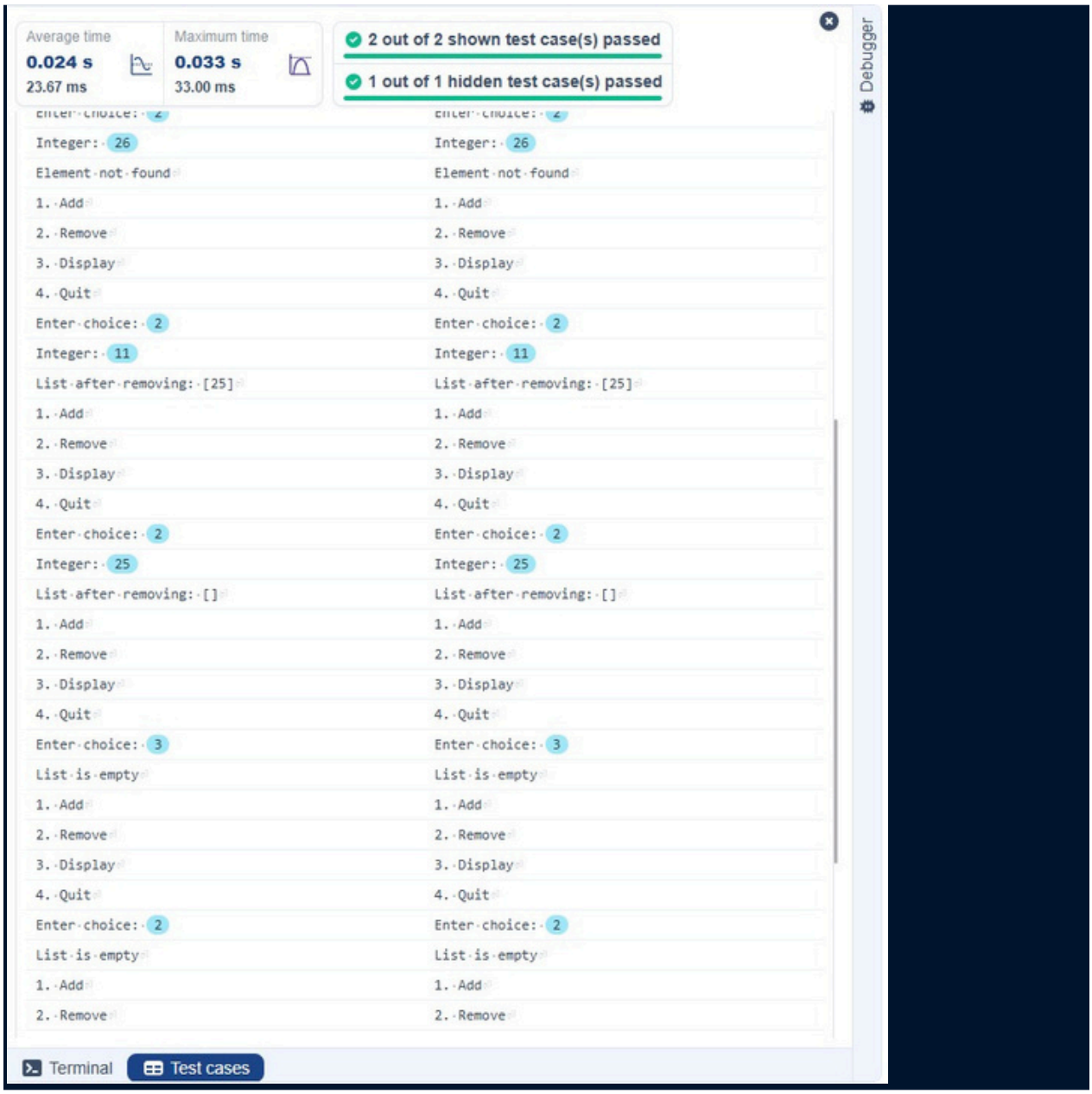
```
1.·Add⏎
2.·Remove⏎
3.·Display⏎
4.·Quit⏎
Enter·choice:·4
```

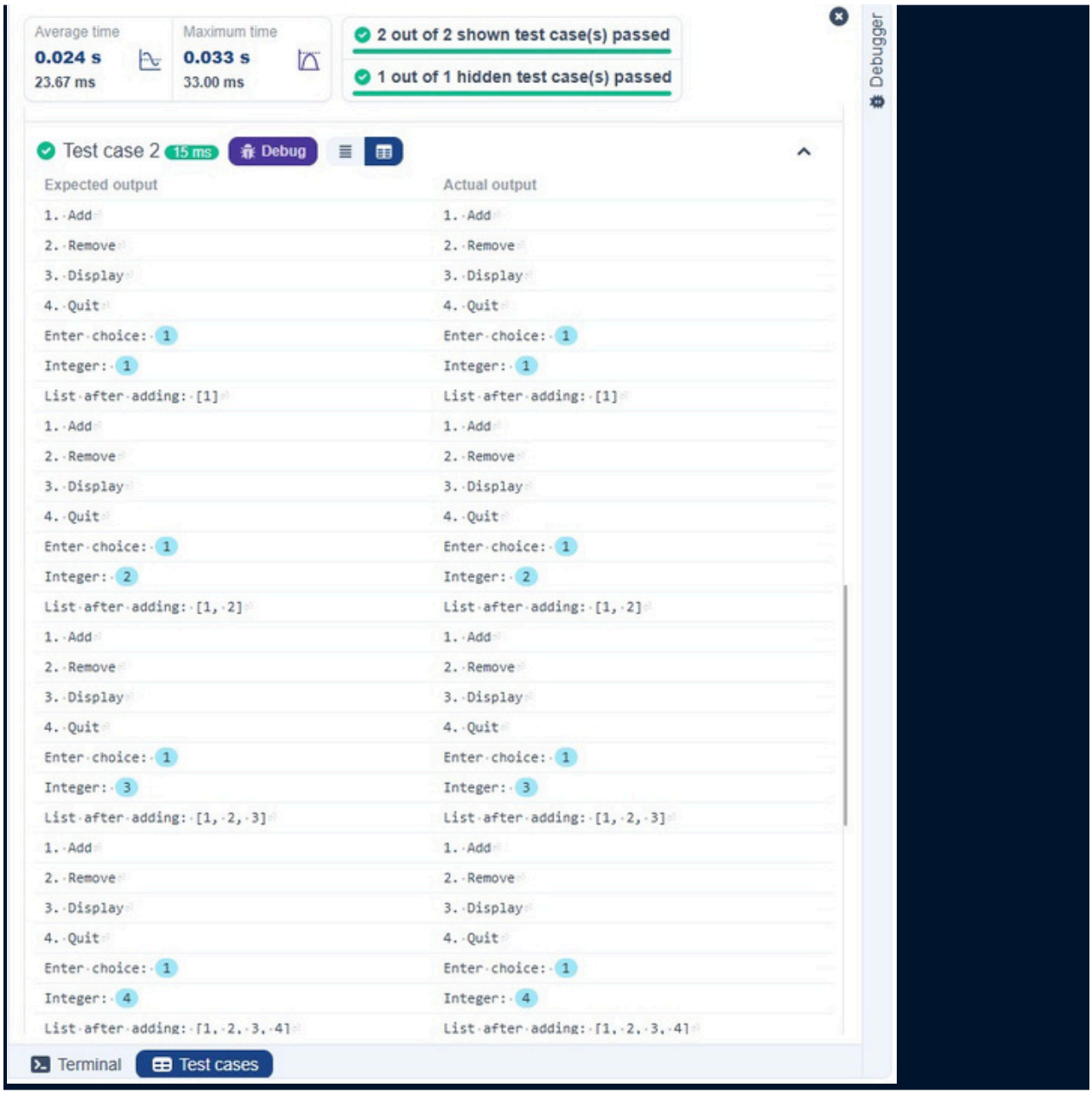
Program code :

```
list=[]
while True:


print("1. Add")
print("2. Remove")
print("3. Display")
print("4.                    Quit")
n=int(input("Enter choice: "))
if n==1:
i=int(input("Integer:              "))
list.append(i)
print(f"List after adding: {list}")
elif n==2:
if list==[]:
print("List is empty")
else:
i=int(input("Integer:              "))
val=True
for j in list:
```

```
if j==i:
list.remove(j)
print(f"List after removing: {list}")
val=False
break
if val:
print("Element not found"

elif n==3:
if list==[]:
print("List is empty")
else:
print(list)
elif n==4:
break
else:
print("Invalid choice")
```


Average time
0.024 s
23.67 ms
Maximum time
0.033 s
33.00 ms
2 out of 2 shown test case(s) passed
1 out of 1 hidden test case(s) passed
Debugger
Test case 1 23 ms Debug
Expected output
Actual output
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 1
Integer: 11
List after adding: [11]
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 1
Integer: 25
List after adding: [11, 25]
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 2
Integer: 26
Element not found
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 2
Integer: 11
List after removing: [25]
Terminal
Test cases


Average time
0.024 s
23.67 ms
Maximum time
0.033 s
33.00 ms
2 out of 2 shown test case(s) passed
1 out of 1 hidden test case(s) passed
Debugger
Integer: 26
Element not found
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 2
Integer: 11
List after removing: [25]
Integer: 25
List after removing: []
Enter choice: 3
List is empty
Terminal
Test cases


Average time
0.024 s
23.67 ms
Maximum time
0.033 s
33.00 ms
2 out of 2 shown test case(s) passed
1 out of 1 hidden test case(s) passed
Debugger
Test case 2 15 ms Debug
Expected output
Actual output
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 1
Integer: 1
List after adding: [1]
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 1
Integer: 2
List after adding: [1, 2]
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 1
Integer: 3
List after adding: [1, 2, 3]
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 1
Integer: 4
List after adding: [1, 2, 3, 4]
Terminal
Test cases


Average time
0.024 s
23.67 ms
Maximum time
0.033 s
33.00 ms
2 out of 2 shown test case(s) passed
1 out of 1 hidden test case(s) passed
Debugger
4. Quit
Enter choice: 1
Integer: 4
List after adding: [1, 2, 3, 4]
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 2
Integer: 2
List after removing: [1, 3, 4]
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 2
Integer: 6
Element not found
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 3
[1, 3, 4]
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 4

2.1.2. Dictionary Operations

11:39

Write a Python program to perform insertion, update, deletion, and traversal operations on a dictionary. An initial dictionary containing 10 predefined records is already given in the program.

Operations to be Performed:
Insertion – Insert a new key-value pair into the dictionary using user input.
Update – Update the value of an existing key using user input.
Deletion – Delete a specified key from the dictionary using user input.
Traversal – Traverse the final dictionary and display all key-value pairs.

Input and Output Format:
Read an integer representing the key to be inserted and a string representing its value. Insert this new key-value pair into the dictionary and display the dictionary after insertion.
Read an integer representing the key to be updated and a string representing the new value. Update the value of the specified key (only if it exists) and display the dictionary after the update.
Read an integer representing the key to be deleted. Delete the specified key from the dictionary (only if it exists) and display the dictionary after deletion.
Finally, traverse the dictionary and display all key-value pairs.
The program should also display the original dictionary before performing any operations.
Each output should be printed with appropriate messages.

Note:
All operations must be performed using dictionary methods.
Perform deletion only if possible; leave the dictionary unchanged.
Refer to the visible test cases for better understanding and strictly match with the input/outputs.

Test case 1

```
Ori gin al ·Di cti on ary :· {1: ·' Ami t' ,·2 :·' R iya ',· 3: ·'K ira n' ,·4 :· 'Ne ha ',· 5:· 'Ar ju n', ·6: ·' Poo ja ',· 7: ·'R ah ul' ,·8 : ·'S neh a' ,·9 :·' Vi kra m' ,·1 0: ·'A njali'}↵
11
Suresh
Aft er· In ser tio n: ·{1 :· 'Am it ',· 2: ·'R iya ' ,·3 :·' Ki ran ',· 4: ·'N eh a', ·5 :·' Arj un' ,· 6:· 'Po oj a', ·7 :·' Ra hul ', ·8: ·'S n eha ',· 9: ·'V ikr am ',· 10 :·' An jal i',·11:·'Suresh'}
↵
3
Karthik
Aft er· Up dat e:· {1 :·' Am it' ,· 2:· 'R iya ',· 3 :·' Kar th ik' ,·4 :· 'Ne ha ',· 5: ·'A rju n', ·6 :·' Poo ja ',· 7: ·'R ah ul' ,· 8:· 'Sn e ha' ,·9 :· 'Vi kra m' ,·1 0: ·'A nj ali ',·11:·'Suresh'}↵
5
Aft er· De let ion :· {1: ·' Ami t' ,·2 :· 'Ri ya' , ·3: ·'K ar thi k', ·4 :·' Ne ha' ,· 6:· 'Po oja ', ·7: ·'R ah ul' ,· 8:· 'S neh a' ,·9 :·' V ikr am' ,· 10: ·'A nj ali ', ·11 :· 'Su resh'}↵
Traversing·Dictionary:↵
1·:·Amit↵
2·:·Riya↵
3·:·Karthik↵
4·:·Neha↵
6·:·Pooja↵
7·:·Rahul↵
8·:·Sneha↵
9·:·Vikram↵
10·:·Anjali↵
11·:·Suresh↵
```

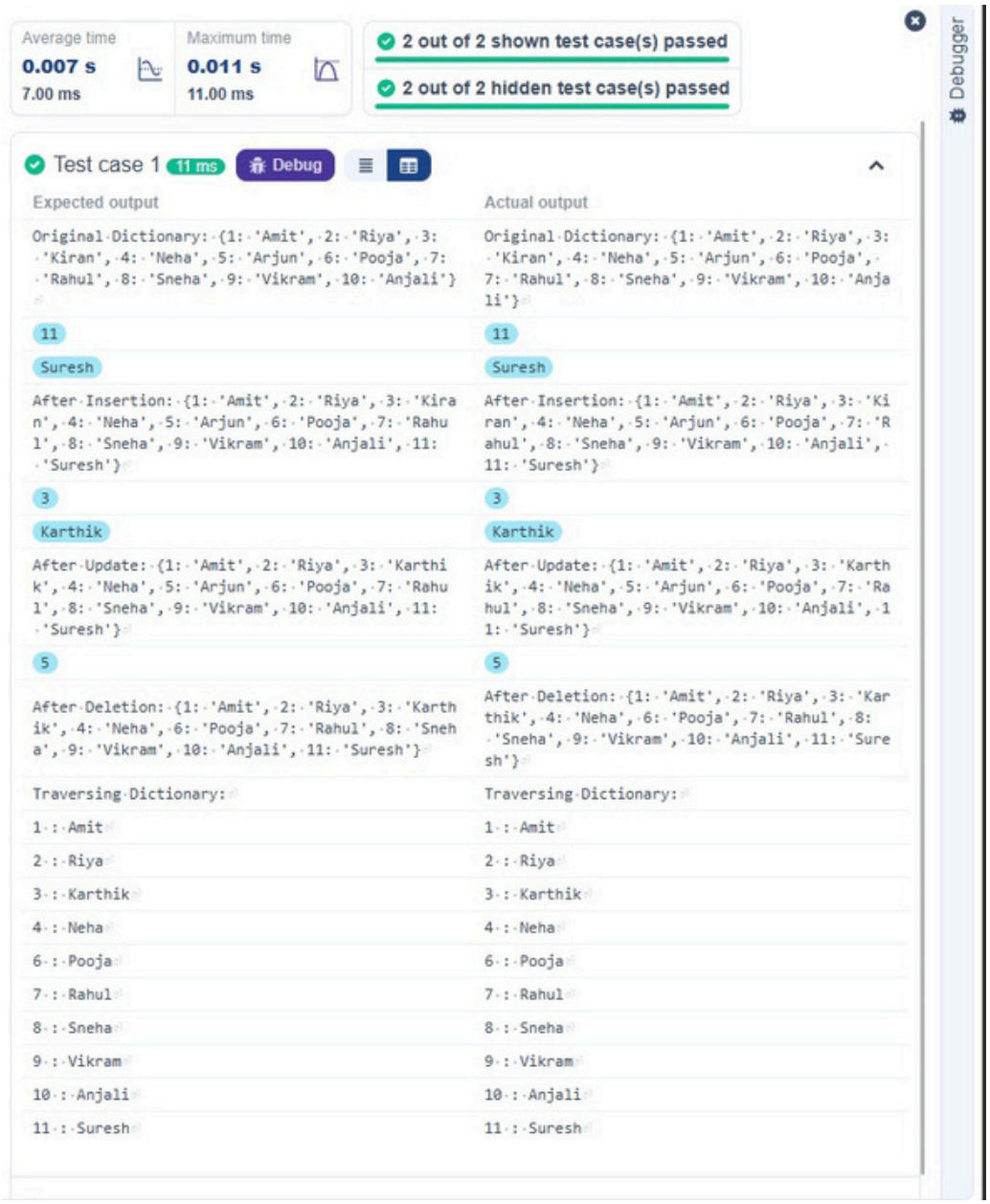
Test case 2

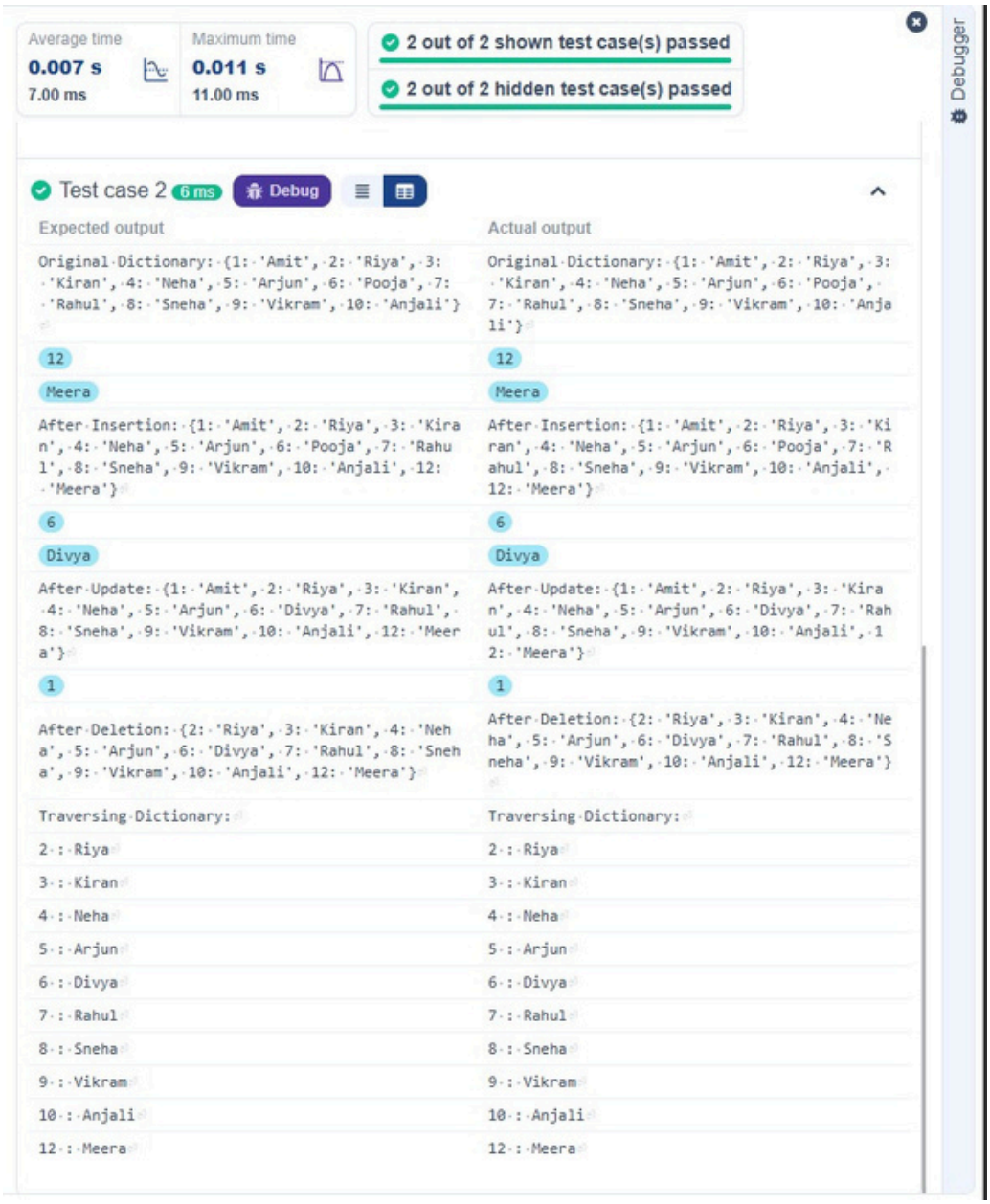
```
Ori gin al ·Di cti on ary :· {1: ·' Ami t' ,·2 :·' R iya ',· 3: ·'K ira n' ,·4 :· 'Ne ha ',· 5:· 'Ar ju n', ·6: ·' Poo ja ',· 7: ·'R ah ul' ,·8 : ·'S neh a' ,·9 :·' Vi kra m' ,·1 0: ·'A njali'}↵
12
Meera
Aft er· In ser tio n: ·{1 :· 'Am it ',· 2: ·'R iya ' ,·3 :·' Ki ran ',· 4: ·'N eh a', ·5 :·' Arj un' ,· 6:· 'Po oj a', ·7 :·' Ra hul ', ·8: ·'S n eha ',· 9: ·'V ikr am ',· 10 :·' An jal i',·12:·'Meera'}↵
6
Divya
Aft er· Up dat e:· {1 :·' Am it' ,· 2:· 'R iya ',· 3 :·' Kir an ',· 4:· 'N eha ', ·5: ·' Arj un' ,·6 :· 'Di vya ', ·7: ·' Rah ul ',· 8: ·'S neh a ',· 9:· 'V ikr am' ,· 10: ·' Anj al i', ·12:·'Meera'}↵
1
Aft er· De let ion :· {2: ·' Riy a' ,·3 :· 'Ki ran ' ,·4 :·' Ne ha' ,·5 :· 'Ar ju n', ·6 :·' Div ya' ,· 7:· 'Ra hu l', ·8 :·' Sn eha ', ·9: ·'V i kra m', ·1 0:· 'An ja li' ,· 12: ·' Mee ra'}↵
Traversing·Dictionary:↵
2·:·Riya↵
3·:·Kiran↵
4·:·Neha↵
5·:·Arjun↵
6·:·Divya↵
7·:·Rahul↵
8·:·Sneha↵
9·:·Vikram↵
10·:·Anjali↵
12·:·Meera↵
```

Program code:

```
student = {
   1: "Amit",
   2: "Riya",
   3: "Kiran",
   4: "Neha",
   5: "Arjun",
   6: "Pooja",
   7: "Rahul",
   8: "Sneha",
   9: "Vikram",
   10: "Anjali"
}
print("Original Dictionary:", student)
key=int(input())
value=input()
student[key]=value
print("After Insertion:",student)
key=int(input())
newvalue=input()
student[key]=newvalue
print("After Update:",student)
key=int(input())
student.pop(key,None)
print("After Deletion:", student)
```

```
print("Traversing Dictionary:")
for key, value in student.items():
print(f"{key} : {value}")
```


Average time
0.007 s
7.00 ms
Maximum time
0.011 s
11.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Debugger
Test case 1
11 ms
Debug
Expected output
Actual output
Original Dictionary: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali'}
Original Dictionary: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali'}
11
11
Suresh
Suresh
After Insertion: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'Suresh'}
After Insertion: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'Suresh'}
3
3
Karthik
Karthik
After Update: {1: 'Amit', 2: 'Riya', 3: 'Karthik', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'Suresh'}
After Update: {1: 'Amit', 2: 'Riya', 3: 'Karthik', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'Suresh'}
5
5
After Deletion: {1: 'Amit', 2: 'Riya', 3: 'Karthik', 4: 'Neha', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'Suresh'}
After Deletion: {1: 'Amit', 2: 'Riya', 3: 'Karthik', 4: 'Neha', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'Suresh'}
Traversing Dictionary:
Traversing Dictionary:
1 : Amit
1 : Amit
2 : Riya
2 : Riya
3 : Karthik
3 : Karthik
4 : Neha
4 : Neha
6 : Pooja
6 : Pooja
7 : Rahul
7 : Rahul
8 : Sneha
8 : Sneha
9 : Vikram
9 : Vikram
10 : Anjali
10 : Anjali
11 : Suresh
11 : Suresh


Average time
0.007 s
7.00 ms
Maximum time
0.011 s
11.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Debugger
Test case 2 6 ms Debug
Expected output
Actual output
Original Dictionary: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali'}
12
Meera
After Insertion: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 12: 'Meera'}
6
Divya
After Update: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Divya', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 12: 'Meera'}
1
After Deletion: {2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Divya', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 12: 'Meera'}
Traversing Dictionary:
2 : Riya
3 : Kiran
4 : Neha
5 : Arjun
6 : Divya
7 : Rahul
8 : Sneha
9 : Vikram
10 : Anjali
12 : Meera

2.2.1. Linear search Technique

04:06

Write a program to check whether the given element is present or not in the array of elements using linear search.

Input format:
The first line of input contains the array of integers which are separated by space
The last line of input contains the key element to be searched

Output format:
If the element is found, print the index. If the element found multiple times, print the starting index
If the element is not found, print Not found.

Sample Test Case:
Input:
1 2 3 4 3 5 6
3
Output:
2

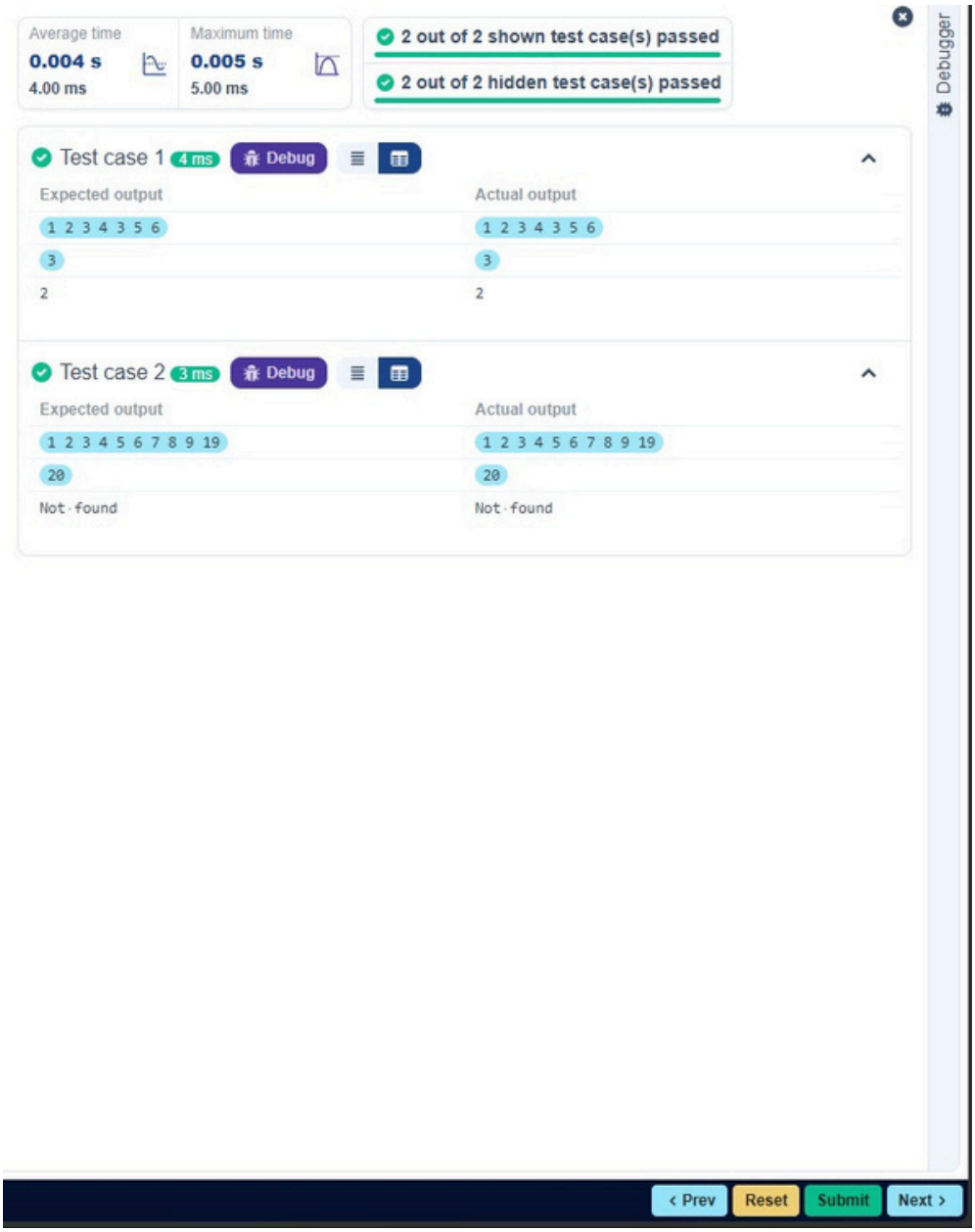
Test case 1

1 2 3 4 3 5 6	
3	
2	

Test case 2

1 2 3 4 5 6 7 8 9 19	
20	
Not found	

```
arr = list(map(int,input().split()))
key = int(input())
found = False

for i in range(len(arr)):
if arr[i] == key:
print(i)
found = True
break

ifnot found:
print("Not found")
```


Average time
0.004 s
4.00 ms
Maximum time
0.005 s
5.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Debugger
Test case 1
4 ms
Debug
Expected output
Actual output
1 2 3 4 3 5 6
3
2
Test case 2
3 ms
Debug
Expected output
Actual output
1 2 3 4 5 6 7 8 9 19
20
Not found
< Prev
Reset
Submit
Next >

2.2.2. Captain of the Team

01:32

You are provided with the heights of 11 cricket players (in centimeters). Your task is to identify the tallest player, who will be selected as the captain of the team.

Input Format:
The first line of input will contain 11 integers, each representing the height of a player (in centimeters), each separated by a space.

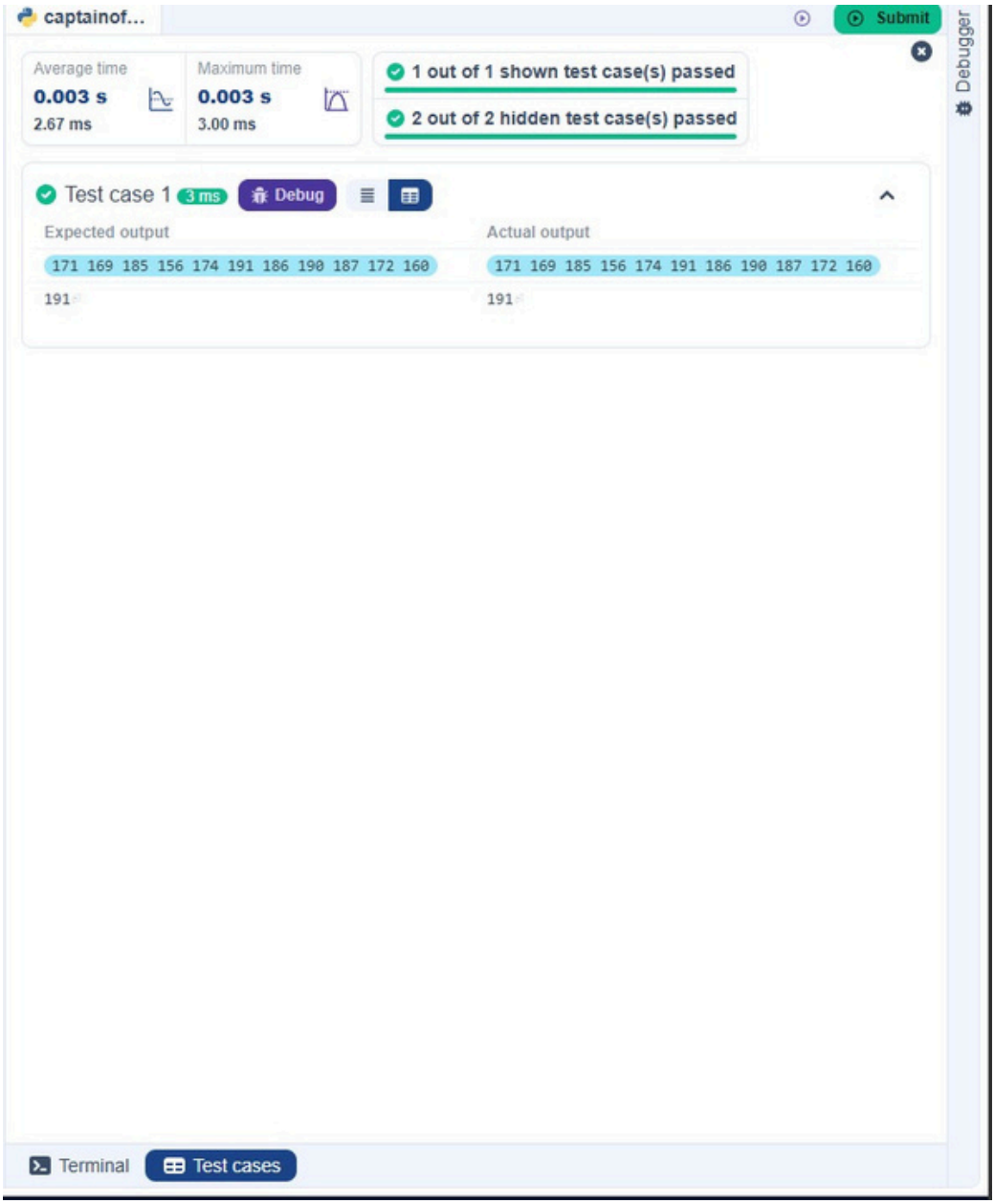
Output Format
The output should be the height (in centimeters) of the tallest player.

Test case 1

```
171 169 185 156 174 191 186 190 187 172 160
191
```

Program code :

```
heights = list(map(int,input().split()))

print(max(heights))
```


captainof...
Submit
Debugger
Average time
0.003 s
2.67 ms
Maximum time
0.003 s
3.00 ms
1 out of 1 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
3 ms
Debug
Expected output
171 169 185 156 174 191 186 190 187 172 160
191
Actual output
171 169 185 156 174 191 186 190 187 172 160
191
Terminal
Test cases